PREMIUM STUDY NOTES

HTML TUTORIAL

A complete, beautifully organized reference covering HTML structure, elements, attributes, formatting, styling, links, media, tables, lists, forms, frames and colors.

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1

Core Foundations

Elements, tags, attributes, nesting rules and document structure — explained from first principles.

2

Styling & Formatting

Text formatting tags, CSS styling methods, colors, and the deprecated tags to avoid.

3

Media, Tables & Forms

Links, images, tables, lists, forms, iframes and a full quick-reference cheat sheet.

"*Good HTML focuses on structure and meaning, not appearance — let the browser handle the display.*"

HTML BASICS · ELEMENTS · FORMATTING · STYLING · LINKS · MEDIA · TABLES · LISTS · FORMS · COLORS

1 · HTML Basics

What is HTML?

HTML is a language for describing web pages.

- **HTML** stands for **Hyper Text Markup Language**
- HTML is not a programming language — it is a **markup language**
- A markup language is a set of **markup tags**
- HTML uses markup tags to describe web pages

```
<html>
  <h1> My First Heading </h1>
  <p> My first paragraph. </p>
</html>
```

HTML Tags

- HTML tags are keywords surrounded by angle brackets, like `<html>`
- Tags normally come in pairs, like `<b>` and `</b>`
- The first tag in a pair is the **start tag** (opening tag); the second is the **end tag** (closing tag)

> Structure: `<p>` Content `</p>` — start tag, content, end tag.

HTML Documents = Web Pages

- HTML documents describe web pages and contain HTML tags plus plain text
- HTML documents are also called web pages

> The purpose of a web browser is to read HTML documents and display them as web pages. The browser does not display the tags themselves — it uses them to interpret the page.

HTML CODE

```
<html>
<body>

  <h1>My First Heading</h1>
  <p>My first paragraph.</p>

</body>
</html>
```

BROWSER OUTPUT

My First Heading

My first paragraph.

Example Explained

- The text between `<html>` and `</html>` describes the web page
- The text between `<body>` and `</body>` is the visible page content
- The text between `<h1>` and `</h1>` is displayed as a heading
- The text between `<p>` and `</p>` is displayed as a paragraph

Editing HTML

HTML can be written and edited with many different editors, such as Visual Studio Code, Sublime Text, or even a plain text editor like Notepad. Using a plain text editor is often recommended when first learning HTML, since it forces you to understand the raw markup.

.htm or .html File Extension?

When you save an HTML file, you can use either the `.htm` or `.html` extension. There is no difference — it's entirely up to you.

2 · Headings, Paragraphs & Elements

HTML Headings

HTML headings are defined with the `<h1>` to `<h6>` tags. `<h1>` defines the most important heading; `<h6>` the least important.

```
<h1>This is a heading</h1>
<h2>This is a heading</h2>
<h3>This is a heading</h3>
```

This is a heading (h1)

This is a heading (h2)

This is a heading (h3)

This is a heading (h4)

This is a heading (h5)

This is a heading (h6)

Browsers automatically add empty space (margin) before and after each heading. Use headings for structure only — never to make text big or bold. Search engines use headings to index page structure, so use `<h1>` as the main heading, followed by `<h2>`, then `<h3>`, and so on.

HTML Paragraphs

HTML paragraphs are defined with the `<p>` tag.

```
<p>This is a paragraph.</p>
<p>This is another paragraph.</p>
```

The browser automatically adds some space (margin) before and after each paragraph.

HTML Elements

An HTML element is everything from the start tag to the end tag.

Start tag	Element content	End tag
`<p>`	This is a paragraph	`</p>`
`<a href="default.htm">`	This is a link	`</a>`
` `	—	—

HTML Element Syntax

- An element starts with a start tag / opening tag
- An element ends with an end tag / closing tag
- The element content is everything between the start and end tag
- Some elements have empty content, and empty elements are closed in the start tag
- Most elements can have attributes

Nested HTML Elements

Most HTML elements can be nested (they can contain other elements). HTML documents consist of nested HTML elements.

```
<html>
  <body>
    <p>This is my first paragraph.</p>
  </body>
</html>
```

This example contains 3 HTML elements: the `<p>` element (defines a paragraph), the `<body>` element (defines the body — its content is another element), and the `<html>` element (defines the whole document — its content is the body element).

Don't Forget the End Tag

Some HTML elements will display correctly even if you forget the end tag:

```
<p>This is a paragraph
<p>This is a paragraph
```

This works in most browsers, because the closing tag is considered optional — but never rely on it. Many elements will produce unexpected results or errors without their end tag, and future versions of HTML may not allow skipped end tags.

Empty HTML Elements

HTML elements with no content are called **empty elements**. `<br>` is an empty element that defines a line break — it has no closing tag.

In XHTML/XML, all elements must be closed. Adding a slash inside the start tag, like `<br />`, is the proper way to close empty elements.

HTML Tip: Use Lowercase Tags

HTML tags are not case sensitive: `<P>` means the same as `<p>`. Many older sites use uppercase tags, but lowercase is the modern convention.

3 · HTML Attributes

What Are Attributes?

- HTML elements can have attributes
- Attributes provide additional information about an element
- Attributes are always specified in the start tag
- Attributes come in name/value pairs like: `name="value"`

Attribute Example

HTML links are defined with the `<a>` tag. The link address is specified in the `href` attribute:

```
<a href="http://www.example.com">This is a link</a>
```

Always Quote Attribute Values

Attribute values should always be enclosed in quotes. Double quotes are most common, but single quotes are also allowed.

Style	Example
Double quotes (common)	`<a href="https://example.com">Link</a>`
Single quotes (allowed)	`<a href='https://example.com'>Link</a>`
Value containing quotes	`<p title='John "ShotGun" Nelson'>Info</p>`

Use Lowercase Attributes

Attribute names and values are case-insensitive, but newer versions of (X)HTML require lowercase attributes.

Good: `<img src="image.jpg" alt="Image">`

Avoid: `<img SRC="image.jpg" ALT="Image">`

Standard Attributes Reference

Attribute	Value	Description
class	classname	Specifies a classname for an element
id	id	Specifies a unique id for an element
style	style_definition	Specifies an inline style for an element
title	tooltip_text	Specifies extra information, shown as a tooltip

HTML Lines & Comments

The `<hr />` tag creates a horizontal line, often used to separate content sections.

```
<p>This is a paragraph</p>
<hr />
<p>This is another paragraph</p>
```

Comments can be inserted into HTML code to make it more readable. Comments are ignored by the browser and are not displayed. There's an exclamation point after the opening bracket, but not before the closing bracket.

```
<!-- This is a comment -->
<p>This is a paragraph.</p>
<!-- Another comment -->
```

Viewing HTML Source

To see how a page was built, right-click anywhere on the page and select "View Page Source" (or similar, depending on the browser). This opens a window containing the raw HTML code of the page.

Core Tag Reference

Tag	Description
`<html>`	Defines an HTML document
`<body>`	Defines the document's body
`<h1>` to `<h6>`	Defines HTML headings
`<hr />`	Defines a horizontal line
`<!-- -->`	Defines a comment

Extra Spaces and Extra Lines

The browser removes extra spaces and extra lines when the page is displayed. Any number of lines counts as one line, and any number of spaces counts as one space. Good HTML focuses on structure and meaning, not appearance — let the browser handle the display.

4 · Text Formatting

Basic Formatting

```
<b>This text is bold</b>
<i>This text is italic</i>
```

This text is bold
This text is italic

`<strong>` usually renders as bold and `<em>` as italic — but they carry semantic meaning ("important" / "emphasized") rather than pure visual style, unlike `<b>` and `<i>`.

Formatting Tags Reference

Tag	Description	Example
`<b>`	Bold text	**This is bold**
`<em>`	Emphasized text	*This is emphasized*
`<i>`	Italic text	*This is italic*
`<small>`	Small text	This is small
`<strong>`	Strong (important) text	**This is strong**
`<sub>`	Subscripted text	H_2O
`<sup>`	Superscripted text	X^2
`<ins>`	Inserted text	Inserted text
`<del>`	Deleted text	~~Deleted text~~

"Computer Output" Tags

Tag	Description	Example
`<code>`	Computer code text	`int x = 10;`
`<kbd>`	Keyboard text	Press `Ctrl + C`
`<samp>`	Sample computer output	`File not found.`
`<var>`	Defines a variable	The value of *x* is 10
`<pre>`	Preformatted text (preserves spacing)	Line 1 / Line 2 / Line 3

Citations, Quotations & Definitions

Tag	Description
`<abbr>`	Defines an abbreviation
`<address>`	Defines contact information for the author/owner of a document
`<blockquote>`	Defines a long quotation, displayed as an indented block
`<q>`	Defines a short quotation, wrapped in quote marks automatically
`<cite>`	Defines the title of a creative work being cited
`<dfn>`	Defines a definition term

Deprecated / avoid: `<font>`, `<center>`, `<s>` / `<strike>`, and `<u>` are deprecated in modern HTML. Use CSS instead — see the Styling chapter.

5 · Styling HTML with CSS

HTML Styles — CSS

CSS (Cascading Style Sheets) is used to style HTML elements — controlling fonts, colors, sizing, spacing and layout. CSS was introduced together with HTML 4 to provide a much better way to style HTML than the old presentational tags.

CSS can be added to HTML in three ways:

- In separate style sheet files (external CSS)
- In a `<style>` element inside the document's `<head>`
- Inline, via the `style` attribute on a single element

Using the Style Attribute

```
<p style="font-family: Verdana; color: red;">
  This is a paragraph styled using the style attribute.
</p>
```

Inline styling is convenient for quick examples, but the **preferred** approach for real projects is to put CSS rules in separate `.css` files.

Background Color

```
<body style="background-color:yellow;">
  <h2 style="background-color:red;">This is a heading</h2>
  <p style="background-color:green;">This is a paragraph.</p>
</body>
```



This is a paragraph.

The `background-color` property makes the old `bgcolor` attribute obsolete.

Font, Color and Size

```
<h1 style="font-family:verdana;">A heading</h1>
<p style="font-family:arial; color:red; font-size:20px;">A paragraph.</p>
```

The `font-family`, `color`, and `font-size` properties make the old `<font>` tag obsolete.

Text Alignment

```
<h1 style="text-align:center;">Center-aligned heading</h1>
```

The `text-align` property makes the old `<center>` tag obsolete.

Deprecated Tags & Attributes

In HTML 4, several tags and attributes were deprecated, meaning they will not be supported in future versions of HTML. The message is clear: **avoid using deprecated tags and attributes** — use CSS styles instead.

Tag / Attribute	Description
`<center>`	Deprecated. Defines centered content — use `text-align`
`<font>`, `<basefont>`	Deprecated. Defines fonts — use `font-family`, `color`, `font-size`
`<s>`, `<strike>`	Deprecated. Strikethrough text
`<u>`	Deprecated. Underlined text
`align`	Deprecated. Alignment of text
`bgcolor`	Deprecated. Background color
`color`	Deprecated. Text color

6 · Links & Images

HTML Hyperlinks (Links)

A hyperlink is a word, group of words, or image you can click to jump to a new document or a section within the current document. When you move the cursor over a link, the arrow turns into a little hand. Links are specified using the `<a>` tag.

Link Syntax

```
<a href="url">Link text</a>
```

> The "link text" doesn't have to be plain text — it can be an image or any other HTML element.

The target Attribute

The `target` attribute specifies where to open the linked document.

```
<a href="https://example.com/" target="_blank">Open in a new tab</a>
```

The id / name Attribute — Bookmarks

The `id` (modern) or `name` (older) attribute can create a bookmark inside a document. Bookmarks are invisible — they don't display specially to the reader.

```
// Create a bookmark named "tips"
<a id="tips">Useful Tips Section</a>

// Jump to it from inside the same page
<a href="#tips">Visit the Tips Section</a>
```

HTML Images

Images are defined with the empty `<img>` tag — it contains attributes only and has no closing tag.

```
<img src="url" alt="some_text" />
```

Attribute	Meaning
`src`	Specifies the path (URL) of the image to display
`alt`	Provides alternate text for the image
`width`	Specifies the width in pixels
`height`	Specifies the height in pixels

The alt Attribute

The required `alt` attribute specifies alternate text for an image if it cannot be displayed.

Situation	What happens
Image cannot be displayed	ALT text is shown instead of the image
Slow connection	ALT text appears until the image loads
Screen reader users	ALT text is read aloud

Set Height and Width

```
<img src="pulpit.jpg" alt="Pulpit rock" width="304" height="228" />
```

> It's good practice to always specify both `height` and `width` — this reserves the space while the page loads, preventing layout shift.

Image & Link Tag Reference

Tag	Description
`<a>`	Defines an anchor (used for both links and bookmarks)
`<img>`	Defines an image
`<map>`	Defines an image-map
`<area>`	Defines a clickable area inside an image-map

7 · Tables & Lists

HTML Tables

Tables are defined with the `<table>` tag. A table is divided into rows with `<tr>`, and each row into data cells with `<td>` ("table data"). Header cells use `<th>` and are displayed bold and centered by default.

```
<table border="1">
  <tr>
    <th>Header 1</th>
    <th>Header 2</th>
  </tr>
  <tr>
    <td>row 1, cell 1</td>
    <td>row 1, cell 2</td>
  </tr>
</table>
```

Header 1	Header 2
row 1, cell 1	row 1, cell 2

> If you don't specify a `border` attribute, the table displays without visible borders — sometimes useful, but usually you'll want borders shown.

Table Key Concepts

Concept	What it does
`colspan` / `rowspan`	Makes a cell span more than one column/row
Nested tags	Elements like `<b>` can be placed inside a `<td>`
`cellpadding`	White space between cell content and its border
`cellspacing`	Distance between cells
`<caption>`	Adds a caption/title to a table
`<thead>/<tbody>/<tfoot>`	Groups header, body and footer content

Table Tag Reference

Tag	Description
`<table>`	Defines a table
`<th>`	Defines a table header cell
`<tr>`	Defines a table row
`<td>`	Defines a table data cell
`<caption>`	Defines a table caption
`<colgroup>` / `<col />`	Groups & formats table columns
`<thead>` / `<tbody>` / `<tfoot>`	Groups header/body/footer content

HTML Lists

Unordered list — `<ul>`

```
<ul>
  <li>Coffee</li>
  <li>Milk</li>
</ul>
```

- Coffee
- Milk

Ordered list — `<ol>`

```
<ol>
  <li>Coffee</li>
  <li>Milk</li>
</ol>
```

1. Coffee
2. Milk

Each item in an unordered list starts with `<li>` and is marked with a bullet; each item in an ordered list is automatically numbered.

Definition Lists

A definition list is a list of items with a description for each. The `<dl>` tag defines the list, used together with `<dt>` (the term) and `<dd>` (its description).

```
<dl>
  <dt>Coffee</dt>
  <dd>- black hot drink</dd>
  <dt>Milk</dt>
  <dd>- white cold drink</dd>
</dl>
```

List Tag Reference

Tag	Description
`<ol>`	Defines an ordered list
`<ul>`	Defines an unordered list
`<li>`	Defines a list item
`<dl>`	Defines a definition list
`<dt>`	Defines a term in a definition list
`<dd>`	Defines the description of a term

8 · Forms & Input

HTML Forms

Forms are used to collect different kinds of user input and pass data to a server. A form can contain input fields, checkboxes, radio buttons, submit buttons and more, and is created with the `<form>` tag.

```
<form>
  input elements
</form>
```

The Input Element

The most important form element is `<input>`, used to select user information. Its behavior varies based on the `type` attribute.

```
<form>
  Name: <input type="text" name="name"><br>
  Password: <input type="password" name="pass"><br>
  <input type="submit" value="Login">
</form>
```

Common Input Types

Type	Description
`text`	Single-line text input field
`password`	Text field with masked (hidden) characters
`radio`	Lets a user select only ONE of a limited set of choices
`checkbox`	Lets a user select ONE or MORE choices
`submit`	Sends form data to the server, per the form's `action`
`reset`	Resets all form fields to default values
`hidden`	An invisible field, still submitted with the form

Radio Buttons & Checkboxes

```
// Radio buttons - only one selectable
<input type="radio" name="sex" value="male"> Male
<input type="radio" name="sex" value="female"> Female

// Checkboxes - multiple selectable
<input type="checkbox" name="vehicle" value="Bike"> I have a bike
<input type="checkbox" name="vehicle" value="Car"> I have a car
```

Form Tag Reference

Tag	Description
`<form>`	Defines an HTML form for user input
`<input />`	Defines an input control
`<textarea>`	Defines a multi-line text input control
`<label>`	Defines a label for an input element
`<fieldset>`	Draws a border around related elements in a form
`<legend>`	Defines a caption for a fieldset
`<select>`	Defines a drop-down list
`<optgroup>`	Groups related options in a select list
`<option>`	Defines an option in a select list
`<button>`	Defines a push button

9 · Frames & Iframes

HTML Frames (Legacy)

With frames, you could display more than one HTML document in the same browser window. Each frame is independent of the others.

> Frames are legacy technology: they are not expected to be supported in future HTML versions, are difficult to use (printing the whole page is hard), and require tracking more documents. Modern layouts should use CSS and iframes instead.

```
<frameset cols="25%,75%">
  <frame src="frame_a.htm" />
  <frame src="frame_b.htm" />
</frameset>
```

> **Important:** you cannot use `<body></body>` together with `<frameset></frameset>`. If you add a `<noframes>` tag for browsers that don't support frames, wrap its text in `<body></body>`.

HTML Iframes (Modern)

An iframe is used to display a web page within a web page.

```
<iframe src="URL"></iframe>
```

- `width` / `height` — size the iframe in pixels (or percent, like "80%")
- `frameborder="0"` — removes the default border
- An iframe can be used as a `target` for a link, by matching the iframe's `name` attribute

```
<iframe src="demo_iframe.htm" name="iframe_a"></iframe>

<a href="https://example.com" target="iframe_a">Load in iframe</a>
```

10 · HTML Colors

Color Values

Colors are displayed by combining RED, GREEN and BLUE light. HTML colors are defined using hexadecimal notation (HEX) for the combination of Red, Green and Blue values (RGB). The lowest value a light source can have is 0 (HEX: `00`); the highest is 255 (HEX: `FF`). HEX values are written as 3 pairs of two-digit numbers, starting with a `#` sign.

Swatch	Color HEX	Color RGB
	#000000	rgb(0,0,0)
	#FF0000	rgb(255,0,0)
	#00FF00	rgb(0,255,0)
	#0000FF	rgb(0,0,255)
	#FFFF00	rgb(255,255,0)
	#00FFFF	rgb(0,255,255)
	#FF00FF	rgb(255,0,255)
	#C0C0C0	rgb(192,192,192)
	#FFFFFF	rgb(255,255,255)

The combination of Red, Green and Blue values from 0–255 gives more than **16 million** different colors (256 × 256 × 256).

Shades of Gray

Gray colors are created by using an equal amount of power for all three light sources:

Swatch	HEX	RGB
	#202020	rgb(32,32,32)
	#404040	rgb(64,64,64)
	#606060	rgb(96,96,96)
	#808080	rgb(128,128,128)
	#A0A0A0	rgb(160,160,160)
	#C0C0C0	rgb(192,192,192)
	#E0E0E0	rgb(224,224,224)

Web Safe Colors — Historical Note

Some years ago, when computers supported a maximum of 256 colors, a list of 216 "Web Safe Colors" was suggested as a web standard. This is not important today, since virtually all computers can display millions of colors.

Named Colors (Selected)

147 color names are defined in the HTML/CSS color specification (17 standard colors plus 130 more). The 17 standard colors are: *aqua, black, blue, fuchsia, gray, grey, green, lime, maroon, navy, olive, purple, red, silver, teal, white,* and *yellow.* A useful selection is shown below:

Swatch	Name	HEX
	Black	#000000
	Blue	#0000FF
	Green	#008000
	Red	#FF0000
	Purple	#800080
	Orange	#FFA500
	Pink	#FFC0CB
	Gold	#FFD700
	Turquoise	#40E0D0
	RoyalBlue	#4169E1
	SeaGreen	#2E8B57
	Brown	#A52A2A
	Gray	#808080
	White	#FFFFFF

11 · HTML Quick Reference

Basic Document Skeleton

```
<html>
<head>
  <title>Title of document goes here</title>
</head>

<body>
  Visible text goes here...
</body>
</html>
```

Headings & Text

```
<h1>Largest Heading</h1> ... <h6>Smallest Heading</h6>
<p>This is a paragraph</p>
<br />  // line break
<hr />  // horizontal rule
<pre>This text is preformatted</pre>
```

Logical & Physical Styles

```
<em>This text is emphasized</em>
<strong>This text is strong</strong>
<code>This is some computer code</code>
<b>This text is bold</b>
<i>This text is italic</i>
```

Links

```
// Ordinary link
<a href="http://www.example.com">Link text goes here</a>

// Image link
<a href="http://www.example.com"><img src="URL" alt="Alternate Text"/></a>

// Mailto link
<a href="mailto:someone@example.com">Send e-mail</a>
```

Lists

```
<ul><li>Item</li><li>Item</li></ul>
<ol><li>First item</li><li>Second item</li></ol>
<dl><dt>Term</dt><dd>Definition</dd></dl>
```

Tables, Iframes & Forms

```
<table border="1"><tr><th>Header</th></tr><tr><td>Cell</td></tr></table>

<iframe src="demo_iframe.htm"></iframe>

<form action="/submit" method="post">
  <input type="text" name="email">
  <input type="submit" value="Send">
</form>
```

Character Entities

Entity	Result
`<`	is the same as <
`>`	is the same as >
`©`	is the same as ©

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